COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10-15-26	**DATE SUBMITTED:**	10-15-26
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ACTIVITY NO. 11 **Containerization**

1. PROCEDURE OUTPUT

1. Create a new repository for this activity.



2. Install Docker and enable the docker socket.

-ubuntu prerequisite

```yaml
tasks:
  - name: Install Ubuntu prerequisites
    ansible.builtin.apt:
      name:
        - ca-certificates
        - curl
      state: present
      update_cache: true
    when: ansible_os_family == "Debian"
```

-docker packages on ubuntu

```yaml
- name: Install Docker packages on Ubuntu
  ansible.builtin.apt:
    name:
      - docker-ce
      - docker-ce-cli
      - containerd.io
      - docker-buildx-plugin
      - docker-compose-plugin
    state: present
    update_cache: true
  when: ansible_os_family == "Debian"
```

-docker packages on centos

```
- name: Install Docker packages on CentOS
  ansible.builtin.dnf:
    name:
      - docker-ce
      - docker-ce-cli
      - containerd.io
      - docker-buildx-plugin
      - docker-compose-plugin
    state: present
  when: ansible_os_family == "RedHat"
```

-enabling and starting of docker

```
- name: Enable and start Docker
  ansible.builtin.systemd:
    name: docker
    enabled: true
    state: started
```

-enabling of docker socket

```
- name: Enable Docker socket
  ansible.builtin.systemd:
    name: docker.socket
    enabled: true
    state: started
```

3. Add to Docker group to your current user.

```
- name: Add SSH user to Docker group
  ansible.builtin.user:
    name: "{{ ansible_user }}"
    groups: docker
    append: true
  when: ansible_user is defined
```

```
masongsong@workstation:~/CPE212_Activity11$ ansible all -i inventory.ini -b -m c
ommand -a "getent group docker" -K
BECOME password:
192.168.56.130 | CHANGED | rc=0 >>
docker:x:982:masongsong
192.168.56.128 | CHANGED | rc=0 >>
docker:x:984:masongsong
masongsong@workstation:~/CPE212_Activity11$
```

4. Create a Dockerfile to install web and DB server.

```
  GNU nano 7.2                          Dockerfile
FROM ubuntu:24.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && apt-get install -y \
    apache2 \
    mariadb-server \
    supervisor \
    && rm -rf /var/lib/apt/lists/*

COPY index.html /var/www/html/index.html
COPY supervisord.conf /etc/supervisor/conf.d/supervisord.conf

RUN mkdir -p /run/mysqld \
    && chown mysql:mysql /run/mysqld \
    && mariadb-install-db --user=mysql --datadir=/var/lib/mysql

EXPOSE 80 3306

CMD ["/usr/bin/supervisord", "-n"]
```

5. Install and build the Dockerfile using Ansible.
-create directory

```
- name: Create application directory
  ansible.builtin.file:
    path: /opt/cpe212-masongact11
    state: directory
    mode: "0755"
```

-copy dockerfile

```
- name: Copy Dockerfile
  ansible.builtin.copy:
    src: Dockerfile
    dest: /opt/cpe212-masongact11/Dockerfile
    mode: "0644"
```

-copy website html

```
- name: Copy website HTML
  ansible.builtin.copy:
    src: index.html
    dest: /opt/cpe212-masongact11/index.html
    mode: "0644"
```

-copy supervisor conf

```
- name: Copy Supervisor configuration
  ansible.builtin.copy:
    src: supervisord.conf
    dest: /opt/cpe212-masongact11/supervisord.conf
    mode: "0644"
```

-build docker image

```
- name: Build Docker image
  ansible.builtin.command:
    cmd: docker build -t cpe212-masongact11:latest /opt/cpe212-masongact11
  register: docker_build
  changed_when: true
```

-remove possible containers

```
- name: Remove existing application container
  ansible.builtin.command:
    cmd: docker rm -f cpe212-activity11
  register: container_removal
  failed_when: false
  changed_when: container_removal.rc == 0
```

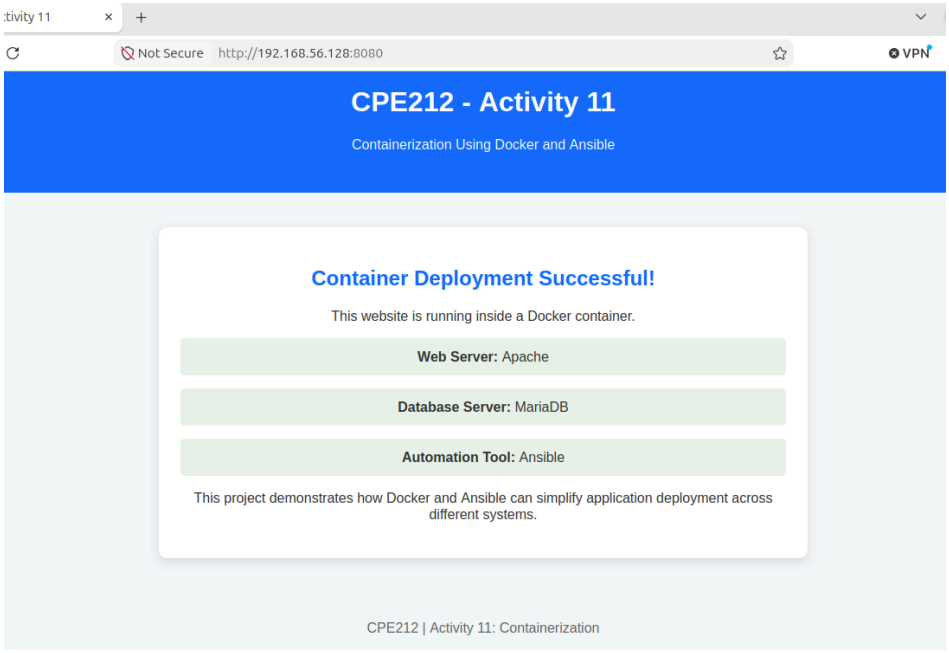
-run container

```yaml
- name: Run application container
  ansible.builtin.command:
    cmd: >-
      docker run -d
      --name cpe212-activity11
      --restart unless-stopped
      -p 8080:80
      -p 3307:3306
      cpe212-masongact11:latest
  register: container_run
  changed_when: true
```

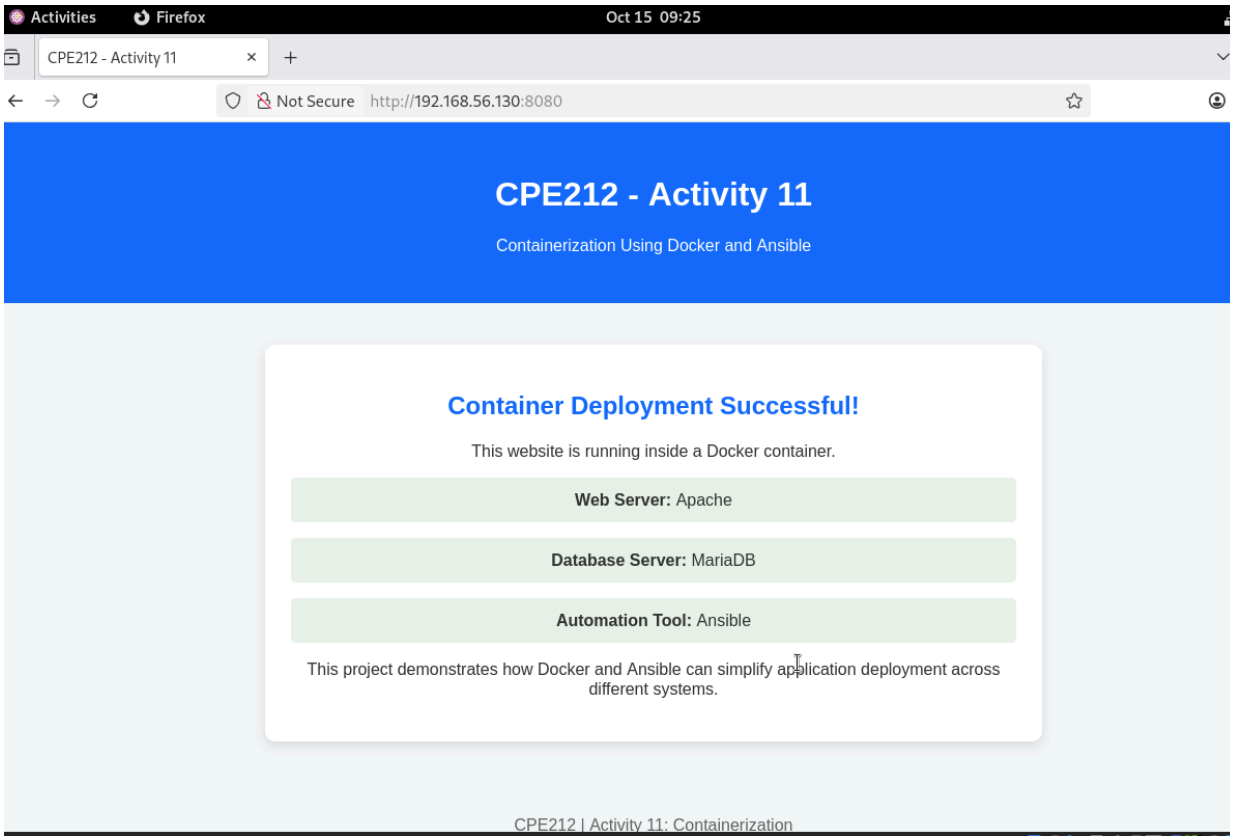
-display output when done

```yaml
- name: Display deployment result
  ansible.builtin.debug:
    msg: "Container deployment completed on {{ inventory_hostname }}."
```

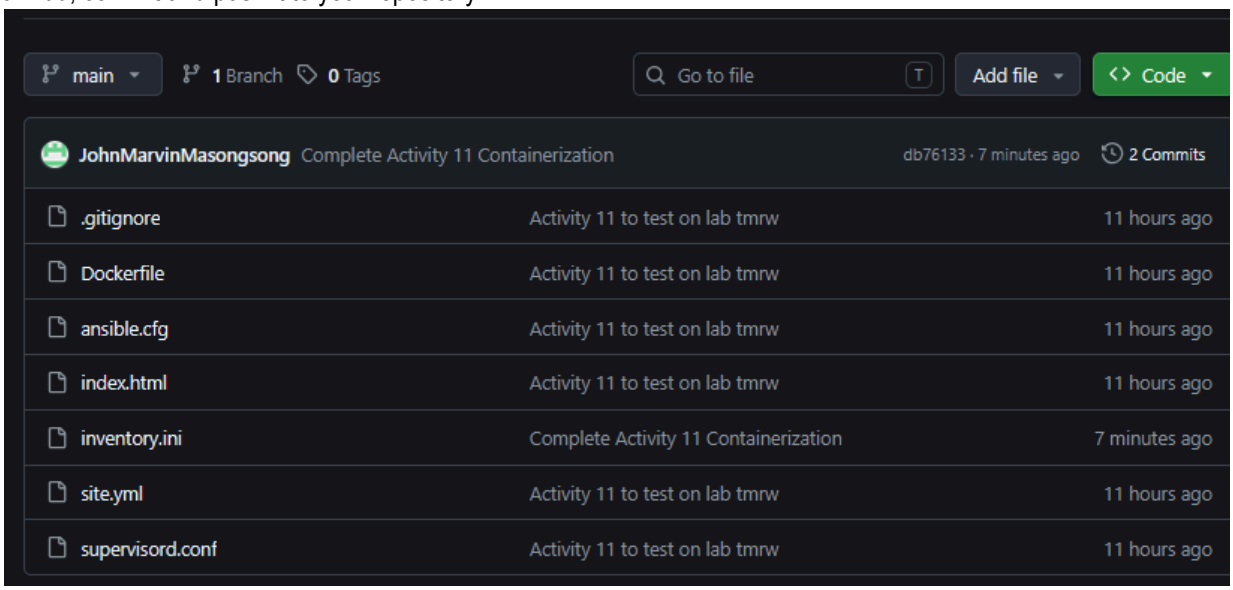
OUTPUT:
Ubuntu



Centos



6. Add, commit and push it to your repository.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:
Answer the following:

1. What are the benefits of implementing containerizations?
   - Besides being efficient and easy to maintain, it is also portable meaning containers can run on different systems and environments as long as they support the container's platform version. It is also isolated meaning applications and its dependencies run in separate containers, which can reduce a lot of load and lessen conflicts.

CONCLUSION:
I have learned a lot from this activity, mainly how to setup dockerfiles, docker images, and setup the website for it while using ansible. It was easy to understand the concept of how each command works since it mainly came from previous HOAs like the ansible builtin commands. All in all we were tasked to download MariaDB and Apache and have it in one container. In conclusion I had fun with this activity, though it was quite hard, I was able to grasp the concept on how to do it.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **NA**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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